THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 9, 4, 1. The top number is: 1

(a) 28 **(b)** 22 **(c)** 14 **(d)** 18

2. Add 48 and its reverse 84. The sum is: 1

(a) 143 **(b)** 121 **(c)** 132 **(d)** 154

3. Kaprekar step: for 5148, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 7074 **(b)** 7173 **(c)** 7083 **(d)** 7092

4. In the hailstone (Collatz) sequence, the number after 7 is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 3

5. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 69: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

6. The digit sum of 70512 is: 1

(a) 8 **(b)** 24 **(c)** 17 **(d)** 15

7. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 49 and 62, the number above them is: 1

(a) 13 **(b)** 121 **(c)** 3038 **(d)** 111

8. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 6, 8, 2. The top number is: 1

(a) 33 **(b)** 32 **(c)** 16 **(d)** 24

9. Add 59 and its reverse 95. The sum is: 1

(a) 165 **(b)** 143 **(c)** 154 **(d)** 176

10. In the hailstone (Collatz) sequence, the number after 15 is: 1

(a) 7 **(b)** 45 **(c)** 47 **(d)** 46

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the hailstone (Collatz) sequence for 10. 2

12. Apply the reverse-and-add process on 78 until a palindrome appears. 2

13. Do one Kaprekar step on 7652. 2

14. Write the hailstone (Collatz) sequence for 12. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the Kaprekar routine on 2194 reaches 6174. 3

16. Find the number of steps taken by the Collatz sequence starting at 20 to reach 1 and the largest number in it. 3

17. A number is formed by writing the digit 2 exactly 12 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

OR

Show that the Kaprekar routine on 5867 reaches 6174. 3

18. Find the number of steps taken by the Collatz sequence starting at 22 to reach 1 and the largest number in it. 3

OR

A number is formed by writing the digit 9 exactly 3 times in a row. Find its digit sum and say whether the number is divisible by 3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take the digits of 5390. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 161 and 272 (both palindromes) gives 433. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 67. 5

20. Take the digits of 4455. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

OR

Show with working that adding 383 and 494 (both palindromes) gives 877. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 79. 5

END OF QUESTION PAPER — Total: 40 Marks

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